WORKED SOLUTION

STEP I 2004, Question 3

Pure • Factors of polynomials

STEP I 2004, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/04-s1-q3

The key idea

A polynomial in two variables can be treated as a polynomial in *one* of them, with coefficients that involve the other. Viewed as a polynomial in x, the factor theorem still works, long division still works, and a quadratic in x still factorises when its discriminant is a perfect square. In part (ii) the leading coefficient in x is itself linear in y, which hints that one factor involves y alone.

Solution

Part (i)

Collect the expression $(*)$ by powers of x:

$$P(x) = x^3 + (2y-5)x^2 + (y^2 - 8y + 6)x + (6y - 3y^2).$$

Substituting $x = 3$,

$$\begin{aligned} P(3) &= 27 + 9(2y-5) + 3(y^2 - 8y + 6) + 6y - 3y^2 \\ &= (27 - 45 + 18) + (18y - 24y + 6y) + (3y^2 - 3y^2) = 0 \end{aligned}$$

for every y, so by the factor theorem $x - 3$ is a factor, as required.

To find the other factor, write $P(x) = (x-3)(x^2 + px + q)$, where p and q may depend on y. Expanding the right-hand side gives $x^3 + (p-3)x^2 + (q - 3p)x - 3q$, and comparing coefficients:

$$\begin{aligned} x^2: &\quad p - 3 = 2y - 5, && \text{so } p = 2y - 2, \\ x^0: &\quad -3q = 6y - 3y^2, && \text{so } q = y^2 - 2y, \end{aligned}$$

and the x coefficient checks out: $q - 3p = y^2 - 8y + 6$, as it should. So the quadratic factor is

$$x^2 + (2y-2)x + y^2 - 2y = (x^2 + 2xy + y^2) - 2(x + y) = (x+y)(x+y-2).$$

Ask yourself How did we spot that the quadratic factorises?

The terms $x^2+2xy+y^2$ are a perfect square $(x+y)^2$, and the remaining terms $-2x-2y$ are a multiple of $x+y$. (If that grouping doesn't jump out, the discriminant in x is $(2y-2)^2-4(y^2-2y)=4$, a perfect square, so the roots are $x=-y+1\pm1$, that is $x=-y$ and $x=2-y$.) Therefore

$$(*)=(x-3)(x+y)(x+y-2),$$

so $a=1$, $b=0$, $c=1$, $d=-2$ (or the same with the two factors swapped: $a=1$, $b=-2$, $c=1$, $d=0$).

Part (ii)

Call the expression Q and again collect by powers of x:

$$Q=(y+2)x^2+(12-4y-5y^2)x+(6y^3-y^2-21y+10).$$

The coefficient of x^2 is $y+2$. If Q is a product of three linear factors, the x^2 term comes from the x-parts of two of them, so it's natural to ask whether $y+2$ divides the whole thing. Set $y=-2$:

$$12+8-20=0,\qquad -48-4+42+10=0,$$

so both remaining coefficients vanish at $y=-2$, and by the factor theorem (in y) each is divisible by $y+2$:

$$\begin{aligned}12-4y-5y^2&=-(5y^2+4y-12)=-(y+2)(5y-6),\\ 6y^3-y^2-21y+10&=(y+2)(6y^2-13y+5).\end{aligned}$$

Also $6y^2-13y+5=(2y-1)(3y-5)$. Hence

$$Q=(y+2)[x^2-(5y-6)x+(2y-1)(3y-5)].$$

Ask yourself Does the bracket factorise, and how can we tell?

Its discriminant in x is

$$\begin{aligned}(5y-6)^2-4(2y-1)(3y-5)&=25y^2-60y+36-(24y^2-52y+20)\\&=y^2-8y+16=(y-4)^2,\end{aligned}$$

a perfect square, so it does. More quickly, $(2y-1)+(3y-5)=5y-6$, so the two numbers with product $(2y-1)(3y-5)$ and sum $5y-6$ are $2y-1$ and $3y-5$, giving roots $x=2y-1$ and $x=3y-5$. Therefore

$$Q=(y+2)(x-2y+1)(x-3y+5).$$

Checks

- Put $x = y = 1$ in part (i): $(*)$ gives $1-5+2+1-8-3+6+6 = 0$, and $(1-3)(2)(0) = 0$. With $x = 2$, $y = 1$: $(*)$ gives $8-20+8+2-16-3+12+6 = -3$, and $(-1)(3)(1) = -3$.
- Put $x = y = 0$ in part (ii): $Q = 10$ and $(2)(1)(5) = 10$. With $x = 1$, $y = 1$: $Q = 6-1-21+2+12-4+1-5+10 = 0$, and indeed $x - 2y + 1 = 0$.
- We checked both factorisations with exact rational arithmetic at 2000 random points (x, y); they agreed every time.

Common slips

- Treating the expression as a polynomial in x but forgetting that the "constant" term is everything without an x, here $6y - 3y^2$ in part (i).
- In part (ii), trying to find a root $x = k$ with k a number: no factor of the form $x - k$ exists. The factor with no x in it, $y + 2$, is the one to find first.
- Sign errors when dividing $-(5y^2 + 4y - 12)$ by $y + 2$; check by expanding.

Beyond STEP

Polynomials in several variables form a ring, and $\mathbb{Z}[x, y]$ can be viewed as $(\mathbb{Z}[y])[x]$: polynomials in x whose coefficients are polynomials in y. That is exactly what we did, and it is why the factor theorem and division still work. These rings have unique factorisation (Gauss's lemma), so the factorisations we found are the only ones up to order and signs. Allenby's *Rings, Fields and Groups* develops these ideas.

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